

TABLEAU

Module 8 - Statistical Analysis in Tableau

Topics in This Module

Topic 1	Reference Lines — Draw a benchmark line on any chart (avg, target, constant)
Topic 2	Trend Analysis — See whether your data is going up, down, or flat over time
Topic 3	Forecasting — Let Tableau predict future values automatically
Topic 4	Box-and-Whisker Plot — See spread, middle value, and outliers of data
Topic 5	Clustering — Let Tableau automatically group similar data points together

Our Practice Dataset — Electronics Shop

Monthly Sales

This is a simple sales dataset from an electronics shop. It shows monthly sales, profit, and units sold across 3 product categories over 18 months. This dataset is perfect for all 5 statistical features in this module.

Dataset: Monthly Sales Data

#	Month	Year	Category	Sales (₹)	Profit (₹)	Units Sold	Region
1	January	2023	Laptops	1,85,000	42,000	12	North
2	February	2023	Laptops	1,72,000	38,000	11	North
3	March	2023	Laptops	2,10,000	51,000	14	North
4	April	2023	Mobiles	98,000	22,000	28	South
5	May	2023	Mobiles	1,12,000	26,000	32	South
6	June	2023	Mobiles	1,35,000	31,000	38	South
7	July	2023	Accessories	45,000	14,000	95	West
8	August	2023	Accessories	52,000	17,000	108	West
9	September	2023	Accessories	61,000	20,000	125	West
10	October	2023	Laptops	2,45,000	62,000	16	North
11	November	2023	Laptops	2,95,000	78,000	19	North
12	December	2023	Laptops	3,20,000	88,000	21	North
13	January	2024	Mobiles	1,25,000	29,000	35	South
14	February	2024	Mobiles	1,18,000	27,000	33	South
15	March	2024	Mobiles	1,52,000	36,000	43	South
16	April	2024	Accessories	71,000	23,000	148	West
17	May	2024	Accessories	68,000	22,000	142	West
18	June	2024	Accessories	79,000	26,000	163	West

Dataset Highlights

Categories: Laptops (high sales, fewer units), Mobiles (medium), Accessories (low sales but many units).

Time range: January 2023 to June 2024 — 18 months of data.

Trend: Sales generally increase over time — good for forecasting and trend analysis.

Topic 1 — Reference Lines



Reference Lines

A straight line drawn on your chart to mark an important value — like an average, a target, or a threshold

1.1 What is a Reference Line?

A Reference Line is just a straight horizontal or vertical line that you draw on your chart at a specific value. It acts like a benchmark or target marker — it helps you instantly see which data points are ABOVE the line and which are BELOW.

Real-life example: *Imagine your teacher draws a line at 50 marks on the class results chart. Every student above the line passed. Every student below failed. That line is a Reference Line. In Tableau, you can draw that kind of line on any chart to mark an average, a target, or a threshold.*

Sales example

You build a bar chart showing monthly sales. You add a Reference Line at the average sales value (₹1,47,000). Now you can instantly see: October, November, December are above average (good months). February and July are below average (weak months). The line makes the comparison obvious without reading every number.

1.2 The 3 Types of Reference Lines

Type	What It Shows	Example in Our Dataset
Constant Line	A fixed number you choose — does not change	Draw a line at ₹2,00,000 as the sales target
Average Line	The average of all values — Tableau calculates it	Average monthly sales = ₹1,47,000 — auto-calculated
Percentile Line	A specific percentile — e.g. top 25% threshold	75th percentile line — marks where top 25% begins

1.3 Where to Find Reference Lines in Tableau

The Analytics Panel

Reference Lines (and all statistical features in this module) live in the Analytics Panel — NOT in the Data Panel. Look at the top-left of your screen. You will see two tabs: 'Data' and 'Analytics'. Click on 'Analytics' to switch. You will see a list of all statistical tools including Reference Lines, Trend Lines, Forecasting, and Clustering.

1.4 Step-by-Step — Add an Average Reference Line

We will build a bar chart of Monthly Sales and add an average line on it.

1	Open a new worksheet in Tableau and connect to the sales dataset
2	Drag 'Month' to the Columns shelf — month names appear as headers
3	Drag 'Sales (₹)' to the Rows shelf — a bar appears for each month
4	Sort by Month chronologically: right-click Month on Columns → Sort → Manual → arrange Jan to Jun in order
5	Now click the 'Analytics' tab at the top of the left panel (next to the 'Data' tab)
6	You will see a list. Under 'Custom', find 'Reference Line'
7	Drag 'Reference Line' and drop it onto your chart A small popup appears with 3 options: Table, Pane, Cell — drop it on 'Table'
8	A dialog box opens. Set these values: → Value: SUM(Sales) → Average → Label: Value (shows the number on the line) → Line style: solid, medium thickness → Colour: choose a dark red or orange so it stands out
9	Click OK — a horizontal line appears across your chart at the average sales value
10	Bars above the line = above average months. Bars below = below average months.
11	To edit the line: right-click it directly on the chart → Edit Reference Line
12	To remove it: right-click → Remove

1.5 Step-by-Step — Add a Constant (Target) Reference Line

Your manager sets a monthly sales target of ₹2,00,000. Let us draw that target line.

1	On the same bar chart from above, go to the Analytics panel again
2	Drag 'Reference Line' → drop on 'Table'
3	In the dialog, change Value from 'Average' to 'Constant'

4	Type the value: 200000 (₹2,00,000)
5	Label: Custom → type 'Sales Target: ₹2,00,000'
6	Change the line colour to green so it looks like a 'target to beat' marker
7	Click OK — a green target line appears at ₹2,00,000
8	Now you can see: only Oct, Nov, Dec 2023 Laptops and some 2024 months beat the target

1.6 Reference Band — Highlighting a Range

A Reference Band is like two Reference Lines with the space between them shaded. It marks a range instead of a single value — e.g. acceptable sales range between ₹1,00,000 and ₹2,00,000.

1	In the Analytics panel, drag 'Reference Band' onto your chart → drop on 'Table'
2	Set the top of the band: SUM(Sales) → 75th Percentile
3	Set the bottom of the band: SUM(Sales) → 25th Percentile
4	Set fill colour to a light yellow with low opacity (around 20–30%)
5	Click OK — a shaded band appears between the 25th and 75th percentile
6	Data points inside the band = normal range. Points outside = unusually high or low.

Topic 2 — Trend Analysis



Trend Analysis

A line that shows the **OVERALL** direction of your data — going up, going down, or staying flat

2.1 What is a Trend Line? — Very Simple Explanation

A Trend Line is a straight (or curved) line that cuts through your data to show the **GENERAL** direction things are moving. Ignore the up-and-down zigzag of individual months. The trend line shows the big picture — is the overall direction upward, downward, or flat?

Real-life example: *Your weight changes every day — some days up, some days down. But if you step back and look at 6 months of data, you can see an overall direction. A trend line on a weight chart would show: you are slowly losing weight overall, even though individual days jump around. Same idea for sales data.*

Sales example

Our dataset shows monthly sales from Jan 2023 to Jun 2024. Some months are higher, some lower. But the trend line will show: overall, sales are **INCREASING** over these 18 months — especially in the Laptops category. This is the kind of insight a manager needs for planning.

2.2 Step-by-Step — Add a Trend Line to a Line Chart

We will build a line chart of monthly sales and add a linear trend line. You have to be create calculated field for date.

1	Open a new blank worksheet
2	Drag 'Month' to the Columns shelf
3	Drag 'Sales (₹)' to the Rows shelf
4	In the Marks card dropdown, change the type from 'Automatic' to 'Line' You now have a line chart showing sales month by month
5	To separate by Category: drag 'Category' to the Color shelf — 3 coloured lines appear (Laptops, Mobiles, Accessories)
6	Now click the 'Analytics' tab on the left panel
7	Under 'Model', find 'Trend Line'

8	Drag 'Trend Line' onto your chart — a small menu appears with trend type options: Linear / Logarithmic / Exponential / Polynomial / Power
9	Drop it on 'Linear' — a straight trend line appears for each category
10	Laptops trend line going steeply upward = strong growth Accessories trend line going moderately upward = slow but steady growth Mobiles trend line = moderate growth
11	Hover over any trend line — a tooltip shows the formula and R-squared value
12	To show only one trend line: right-click the trend line → Edit Trend Lines → uncheck categories you do not want

2.3 Understanding the Trend Line Tooltip — Basics

When you hover over a trend line, Tableau shows a small box with numbers. Here is what they mean in simple terms:

What You See	What It Means in Simple English
Sales = 1500*Month + 50000	The formula: for every extra month, sales go up by about ₹1,500
R-Squared = 0.87	How well the line fits the data. 0.87 = 87% fit = very good. Closer to 1.0 = better fit.
P-value = 0.002	Is this trend real or just random? Less than 0.05 = the trend is REAL and not by chance.

Just Remember These Two Things

R-Squared: the closer to 1.0, the stronger the trend. 0.9+ = very strong. 0.5 = weak.

P-value: if it is less than 0.05, the trend is real. If it is more than 0.05, the trend might just be random noise.

Topic 3 — Forecasting



Forecasting

Tableau looks at your past data and PREDICTS what future values will be

3.1 What is Forecasting? — Very Simple Explanation

Forecasting means making an educated guess about the FUTURE based on what happened in the PAST. Tableau looks at your historical data, finds the pattern, and extends that pattern forward into future months.

Real-life example: *Every year, umbrella sales jump in June-July (monsoon season). A shopkeeper can predict that next June, umbrella sales will also jump. They do not know the exact number but they can forecast a range. Tableau does this automatically for your sales data — it finds seasonal patterns and trends and predicts future values.*

What Tableau Needs to Generate a Forecast

1. A DATE field — Forecasting only works when you have dates (month, year, day) on your chart.
2. At least 5 data points — Tableau needs enough history to find a pattern.
3. A continuous date axis — the date must be shown as a continuous line, not discrete categories.

Our dataset has 18 months of data — perfect for forecasting!

3.2 Understanding the Forecast — 3 Parts

Part of Forecast	What It Looks Like	What It Means
Forecast Line	A dotted/dashed line extending past your data	Tableau's best guess for future values
Confidence Band (shaded area)	A shaded region around the forecast line	The range within which the actual value will likely fall

Simple example: *Tableau might forecast that July 2024 Laptop sales will be ₹2,80,000. But it also shows a confidence band from ₹2,20,000 to ₹3,40,000. This means: most likely ₹2,80,000, but it could be anywhere in that range. The wider the band, the less certain the forecast.*

3.3 Step-by-Step — Generate a Forecast

We will forecast the next 6 months of sales based on our 18-month history.

1	Open a new blank worksheet
2	Drag 'Month' to the Columns shelf Important: make sure Month is set as CONTINUOUS (green pill, not blue) If it shows as blue, right-click the Month pill → Continuous
3	Drag 'Year' to the Columns shelf as well — place it BEFORE Month so the axis reads: 2023 Jan, 2023 Feb... 2024 Jun
4	Drag 'Sales (₹)' to the Rows shelf
5	Change the mark type to 'Line' — you see the 18-month sales trend as a line chart
6	Now click the 'Analytics' tab on the left panel
7	Under 'Model', find 'Forecast'
8	Drag 'Forecast' and drop it onto your chart Tableau immediately adds a forecast extension to your line!
9	A shaded grey or blue area appears beyond your last data point (June 2024) This is the forecast for July 2024 onwards
10	To change forecast settings: go to the top menu → Analysis → Forecast → Forecast Options
11	In Forecast Options you can change: → Forecast Length: how many months forward (set to 6 months) → Ignore Last: skip the most recent months if they are incomplete → Forecast Model: Automatic (let Tableau choose — best for beginners)
12	Click OK — the forecast now shows 6 months into the future
13	Hover over the forecast area — the tooltip shows the predicted value and the confidence range

Topic 4 — Box-and-Whisker Plot



Box-and-Whisker Plot

Shows the SPREAD of data — where is the middle? How wide is the range? Are there any unusual values?

4.1 What is a Box Plot? — Very Simple Explanation

A Box-and-Whisker Plot (or just Box Plot) is a special chart that shows you 5 key numbers about a set of data — all in one picture. It tells you: what is the minimum, what is the maximum, where is the middle, and where do most values sit. It also highlights outliers — values that are unusually high or low.

Real-life example: *Imagine 10 students take a test. Most scored between 60 and 80. One student scored 20 (very low) and one scored 98 (very high). A box plot would show: the middle box covers 60–80 (where most students are), the whiskers extend to 20 and 98, and the two extreme students might appear as dots (outliers). The teacher sees the full picture at a glance.*

4.2 The 5 Numbers a Box Plot Shows

#	Name	Simple Meaning	In Our Sales Data
1	Minimum	The smallest value in the data (bottom whisker tip)	₹45,000 — Accessories Jul 2023
2	Q1 (25th percentile)	25% of data is below this point (bottom of the box)	₹85,500 — 25% of months had sales below this
3	Median	The exact middle value — half above, half below (line inside box)	₹1,28,500 — the true middle monthly sales
4	Q3 (75th percentile)	75% of data is below this point (top of the box)	₹2,00,000 — 75% of months had sales below this
5	Maximum	The largest value (top whisker tip)	₹3,20,000 — Laptops Dec 2023

4.3 What is an Outlier?

Outlier = An Unusually High or Low Value

An outlier is a data point that is far away from all the others — so extreme that it stands out. In a box plot, outliers appear as separate dots above the top whisker or below the bottom whisker.

In our data: December 2023 Laptops sales of ₹3,20,000 might appear as an outlier dot above the box — because it is much higher than a typical month. This was the festival season effect.

Outliers are important! They tell you something unusual happened — a promotion, a supply problem, a big corporate order, or a seasonal spike.

4.4 Step-by-Step — Build a Box-and-Whisker Plot

We will compare the spread of sales values across the 3 categories using a box plot.

1	Open a new blank worksheet
2	Drag 'Category' to the Columns shelf — 3 category names appear: Accessories, Laptops, Mobiles
3	Drag 'Sales (₹)' to the Rows shelf — you see one bar or dot per category
4	Now we need to see individual data points: drag 'Month' to the Detail shelf in the Marks card (This tells Tableau to plot each month as a separate point)
5	In the Marks card dropdown, change the type to 'Circle' You now see individual dots — one dot per month per category
6	Now go to the Analytics tab on the left panel
7	Under 'Summarize', find 'Box Plot'
8	Drag 'Box Plot' and drop it onto your chart The box-and-whisker structure instantly appears over your dots!
9	You will see: → A rectangular box for each category (this is the middle 50% of data) → A horizontal line inside the box (this is the median) → Vertical lines extending above and below (these are the whiskers) → Any dots outside the whiskers (these are the outliers)
10	Colour the boxes: in the Analytics panel, right-click the Box Plot → Format → change fill colour
11	To see exact values: hover over any part of the box plot — the tooltip shows Min, Q1, Median, Q3, Max
12	Title: 'Sales Distribution by Category — Box Plot'

4.5 How to Read the Box Plot — Our Sales Data

Category	Box Width	What It Means	Outliers?
Laptops	Wide box (₹1,72,000 to ₹3,20,000)	Very spread out — sales vary a lot month to month	Yes — Dec 2023 appears as outlier dot above box
Mobiles	Medium box (₹98,000 to ₹1,52,000)	Moderate spread — reasonably consistent	No outliers — all months within normal range
Accessories	Narrow box (₹45,000 to ₹79,000)	Tightest spread — most consistent sales	No outliers — very steady category

Business insight from the box plot: Accessories are the most predictable category (narrow box). Laptops are the most volatile (wide box) — likely due to seasonal demand and festival sales spikes. Mobiles are in the middle.

Topic 5 — Clustering

● Clustering

Tableau automatically groups similar data points together — you do not need to define the groups yourself

5.1 What is Clustering? — Very Simple Explanation

Clustering means grouping similar items together automatically. You give Tableau a set of data points and say 'find groups that are naturally similar'. Tableau uses a mathematical method to identify which points belong together — without you having to manually define the groups.

Real-life example: *Imagine you have 100 customers. You do not know how to group them. You ask Tableau: 'find groups based on how much they spend and how often they buy'. Tableau might find: Group 1 = high spenders who buy rarely (big ticket buyers), Group 2 = low spenders who buy often (frequent small buyers), Group 3 = occasional buyers (mid-range). You did not define these groups — Tableau discovered them from the data.*

Clustering vs Groups (from Module 5)

In Module 5, you created Groups by MANUALLY deciding which items belong together (e.g. Mumbai + Pune = West Zone). Clustering is the AUTOMATIC version — Tableau finds the natural groupings in the data based on patterns and similarity. You just tell it how many groups to create.

5.2 How Clustering Works — No Math Required

Tableau uses a method called K-Means Clustering. Here is the basic idea in simple terms:

- **You tell Tableau:** 'Find 3 clusters in this data'
- **Tableau looks at:** how similar data points are to each other (based on the measures you choose)
- **It assigns:** each point to the cluster it is most similar to
- **Result:** each data point gets a cluster label — Cluster 1, Cluster 2, Cluster 3
- **You then:** analyse each cluster to understand what makes it unique

5.3 Step-by-Step — Add Clustering to a Scatter Plot

We will cluster our months based on Sales and Profit — find natural groups of high performance, medium, and low performance months.

1

Open a new blank worksheet

2	Build a scatter plot first: Drag 'Sales (₹)' to the Columns shelf Drag 'Profit (₹)' to the Rows shelf
3	Drag 'Month' to the Detail shelf on the Marks card (Each month becomes one dot — 18 dots total)
4	Drag 'Category' to the Color shelf to colour the dots by category first (optional — you can skip this)
5	Now click the 'Analytics' tab on the left panel
6	Under 'Model', find 'Cluster'
7	Drag 'Cluster' and drop it onto your chart — a small dialog appears asking how many clusters
8	Set Variables: Tableau may auto-select Sales and Profit. Keep those. Set Number of Clusters: start with 3
9	Click OK — Tableau immediately colours the dots into 3 groups: → Cluster 1 (e.g. blue) = high sales, high profit months → Cluster 2 (e.g. orange) = medium sales and profit months → Cluster 3 (e.g. red) = low sales, low profit months
10	A new field called 'Clusters' appears in the Data panel — you can use it in other charts too
11	Hover over any dot — the tooltip shows which cluster it belongs to
12	To change the number of clusters: right-click 'Clusters' in the Data panel → Edit Clusters → change the number

5.4 What the Clusters Mean in Our Data

Cluster	Typical Months	Sales Range	Profit Range	Business Meaning
Cluster 1 (High)	Oct, Nov, Dec 2023 Laptops	₹2,45,000–₹3,20,000	₹62,000–₹88,000	Festival season peak — best performing months
Cluster 2 (Medium)	Mar 2023 Laptops, Jun 2023 Mobiles	₹1,35,000–₹2,10,000	₹31,000–₹51,000	Normal good months — solid performance
Cluster 3 (Low)	Jul–Sep 2023 Accessories	₹45,000–₹79,000	₹14,000–₹26,000	Accessory months — low revenue, different product mix

Module 8 — Quick Reference Cheat Sheet

All 5 Statistical Features — Summary

Feature	What It Does	Found In	Needs
Reference Line	Draws a line at avg / target / constant value on chart	Analytics Panel → Custom → Reference Line	Any chart with a measure axis
Reference Band	Shades a range between two values on chart	Analytics Panel → Custom → Reference Band	Any chart with a measure axis
Trend Line	Shows overall direction of data (going up/down/flat)	Analytics Panel → Model → Trend Line	At least 5 data points
Forecast	Predicts future values based on past patterns	Analytics Panel → Model → Forecast	A DATE field + at least 5 data points
Box Plot	Shows spread: min, Q1, median, Q3, max, outliers	Analytics Panel → Summarize → Box Plot	Multiple data points per category
Clustering	Automatically groups similar data points together	Analytics Panel → Model → Cluster	A scatter plot or data with 2+ measures

Quick Build Steps

REFERENCE LINE: Build chart → Analytics tab → drag 'Reference Line' → drop on Table

Set Value = Average OR Constant → type value → choose colour → OK

TREND LINE: Build line chart with dates → Analytics tab → drag 'Trend Line' → drop on Linear

Hover over the line to see R-squared and formula

FORECAST: Build line chart with CONTINUOUS date → Analytics tab → drag 'Forecast' → drop on chart

Go to Analysis → Forecast → Forecast Options to set length and model

BOX PLOT: Category → Columns | Sales → Rows | Month → Detail | Marks = Circle

Analytics tab → drag 'Box Plot' → drop on chart

CLUSTERING: Build scatter plot (Sales → Columns, Profit → Rows, Month → Detail)

Analytics tab → drag 'Cluster' → drop on chart → set number of clusters → OK

Reading Statistical Results — Quick Guide

What You See	What It Means
Bar above the reference line	That month/item is above average or above target
Trend line going upward (positive slope)	Overall sales are growing over time
R-Squared close to 1.0	Trend is a strong fit to the data
P-value less than 0.05	Trend is statistically real, not just random noise
Forecast line (dotted) going up	Tableau predicts future growth will continue
Wide confidence band on forecast	Low confidence — future is uncertain, wide range of possibilities
Narrow confidence band	High confidence — future prediction is more reliable
Wide box in box plot	That category has very inconsistent/spread out values
Narrow box in box plot	That category is very consistent/stable
Dot outside the whiskers	Outlier — something unusual happened that month
Cluster 1 = top-right on scatter plot	High performance cluster — both measures are high
Cluster 3 = bottom-left on scatter plot	Low performance cluster — both measures are low

End of Module 8 — Statistical Analysis in Tableau

Practice: Build all 5 features on the sales dataset. Start with Reference Lines (easiest) and finish with Clustering (most advanced).

Next → Module 9: Dashboards in Tableau — Layout, Actions, Filters and Design